

SMART INDIA HACKATHON 2026

Problem Statement ID – SIH26133

Problem Statement Title - Accessibility and quality of public healthcare services, particularly in rural and underserved areas

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID -

Team Name - BugBusters

GraminCare

Offline-First Smart Rural Telehealth Ecosystem

TECHNICAL APPROACH

1. Technologies to be used:

- Frontend: React 18, Vite, Lucide Icons, Modern CSS
- Backend: Node.js, Python, Server-Sent Events (SSE) for sync
- Offline Database: IndexedDB, Service Workers, LocalStorage Sync Queue
- Hardware & Alerts: GPS Geolocation API, Camera/Mic, SMS/USSD & LoRa fallback for emergency SOS

2. Methodology & Implementation Process:

- Offline Screening: ASHA worker records vitals offline; auto-assigns color triage (Red/Yellow/Green) saved in IndexedDB.
- Auto Sync Engine: Background engine detects network and syncs pending records to backend via SSE.
- Doctor Console: PHC doctor receives live high-risk alerts, conducts low-bandwidth WebRTC video call, and issues digital e-prescriptions.
- Emergency SOS: High-risk RED alerts instantly dispatch GPS snapshot & patient vitals to PHC Emergency Room & 108 Ambulance.

FEASIBILITY AND VIABILITY

IMPACT AND BENEFITS

RESEARCH AND REFERENCES

ARCHIVE I: Health Ministry & ABDM

MoHFW Rural Health Statistics (RHS 2022-23)

Documents 1:10,000 doctor ratio & necessity for ASHA triage.

Ayushman Bharat Digital Mission (ABDM Stack)

Technical specs for ABHA ID & Health Repository Provider.

ARCHIVE II: WHO & NITI Aayog Guidelines

WHO Digital Health Interventions Guidelines

Evidence validating offline decision-support apps.

NITI Aayog Telemedicine Practice Guidelines

Protocols for remote consultation & e-prescriptions.

ARCHIVE III: Web Specs & Offline Sync

W3C IndexedDB & Service Worker Specs

Client-side offline database caching & background queueing.

WebRTC Peer-to-Peer Protocol (IETF RFC 8825)

Real-time video & low-bandwidth canvas relay protocols.

ARCHIVE IV: Security & Clinical Protocols

Digital Personal Data Protection Act (DPDP 2023)

Patient health data privacy, encryption & consent compliance.

ICMR High-Risk Pregnancy ANC Triage Protocols

Standard clinical thresholds for maternal vital sign escalation.